COMP0008: Computer Architecture & Concurrency – CW1

Dear students,

The questions below will not be marked and they are for you to test your understanding.

We will publish solutions in 1-2 weeks that will allow you to check your answers.

If any of the questions is not clear, please post a question in the forum or attend the drop-in sessions.

COMP0008 staff.

1. Let $x = 13_4$ and $y = 11_{16}$. How many digits $(x \times y)$ has in base 64?
2. Let x, y be 32-bit integers. How many bits are guaranteed to be enough for representing $(x \times y)$ for any values of x, y?
3. Suppose that x and y both have d digits (without leading zeros) in base b. How many digits will $(x \times y)$ have in base b^2? Give the smallest feasible interval $[l, h]$. (i.e., at least l and at most h).
4. What is the 2's complement of $\mathrm{0xBADC0C0A}$? Calculate it without going through binary.